

Captain Obi 2.0 – AI Powered Autonomous FireFighting Robot

Artificial Intelligence Project Report



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Captain Obi 2.0 – In a Nutshell

Captain Obi 2.0 is an AI-based autonomous firefighting robot designed to detect fire, navigate intelligently toward it, extinguish it automatically, and continue patrolling the environment. The project is an upgraded version of our previous firefighting robot developed in the Introduction to Data Science course. In this enhanced version, Artificial Intelligence concepts such as computer vision, autonomous navigation, path planning, intelligent agents, and wireless communication are integrated to transform the robot from a simple sensor-based system into an intelligent autonomous agent.

The robot uses computer vision and a trained dataset to distinguish real fire from other light sources such as bulbs and sunlight. Communication between the laptop, camera system, and ESP/robot is handled using MQTT protocol for real-time command transfer. The robot operates in both automatic and manual modes. In automatic mode, it patrols the environment, detects fire, calculates the nearest safe route, moves toward the fire, sprays water automatically, and refills water when needed. In manual mode, the user can control the robot using keyboard or controller commands.

A Python/Pygame simulation environment was also developed to visualize and test the robot's autonomous behavior, fire detection, obstacle avoidance, water management, and navigation logic.



Captain Obi 2.0 – Autonomous AI Firefighting Robot

Introduction

Fire hazards remain one of the most dangerous threats in homes, laboratories, offices, and industrial environments. Early detection and rapid response are extremely important in minimizing damage and saving lives. Human intervention during the initial moments of a fire can often be dangerous and delayed.

Captain Obi 2.0 was developed as an intelligent autonomous firefighting robot capable of detecting fire, navigating indoor environments, avoiding obstacles, and extinguishing fire automatically. Unlike traditional firefighting robots that depend only on flame sensors, our upgraded system integrates Artificial Intelligence and computer vision for smarter decision-making and more reliable fire recognition.

The robot combines robotics, AI algorithms, computer vision, autonomous navigation, MQTT communication, and Python simulation into one integrated system. The system demonstrates how low-cost AI-enabled robotic systems can support safety operations and early fire response applications.

Objectives of the Project

- Develop an AI-powered autonomous firefighting robot.
- Implement computer vision-based fire detection.
- Reduce false fire detection caused by bulbs or sunlight.
- Integrate autonomous navigation and obstacle avoidance.
- Implement automatic and manual operating modes.
- Develop a Python/Pygame simulation environment.
- Implement MQTT-based communication between devices.
- Demonstrate AI concepts in a real-world robotics application.

Project Overview

Captain Obi 2.0 is a mobile robotic firefighting system equipped with motors, ESP32, phone camera, laptop AI model, water tank, and water pump mechanism. The robot continuously scans the environment for fire using computer vision and sensor input.

When fire is detected, the robot calculates the nearest safe path toward the fire source, moves autonomously while avoiding obstacles, stops at a safe distance, and activates the water pump to extinguish the fire. After extinguishing the fire, the robot resumes search mode and continues patrolling the environment.

The system also includes a water management mechanism. If the water tank becomes low, the robot automatically navigates to a refill station, refills its tank, and continues operation.

The project also includes a virtual simulation environment developed in Python using Pygame, where fires appear dynamically and the robot performs autonomous firefighting operations visually.

Key Features

- 1. AI-Based Fire Detection**
Uses computer vision and trained datasets to recognize real fire.
- 2. Autonomous Navigation**
Robot automatically moves toward the nearest detected fire.
- 3. Automatic Water Spray System**
Robot stops at a safe distance and sprays water automatically.
- 4. Water Refill System**
Robot automatically moves to the refill station when water becomes low.
- 5. Manual and Automatic Modes**
User can switch between autonomous mode and manual control mode.
- 6. Python/Pygame Simulation**
A complete simulation environment for testing robot behavior and AI logic.

7. MQTT Communication

Real-time communication between camera system, laptop, and robot controller.

Artificial Intelligence Modules

1. Computer Vision

The robot uses computer vision to detect fire using a live camera feed instead of depending only on traditional flame sensors. A trained dataset containing images of:

- Real fire
- Non-fire objects
- Bulb/light conditions
- Daylight environments
- Obstacles

was used to improve fire classification accuracy and reduce false alarms.

The laptop processes live video frames and identifies fire in real time before sending commands to the robot.

2. UDP Socket Communication

UDP socket communication is used for real-time wireless communication between the laptop, camera system, and ESP32 controller.

The AI model processes the camera feed on the laptop and sends movement commands such as:

- Move Forward
- Turn Left
- Turn Right
- Stop
- Spray Water

This communication method allows fast data transfer with low delay for autonomous robot operation.

4. Automatic and Manual Modes

The system supports both autonomous and manual operation modes.

Automatic Mode

- Robot patrols automatically
- Detects fire
- Moves toward target

- Sprays water automatically
- Refills water automatically
- Resumes search mode

Manual Mode

- User controls robot using keyboard/controller
- User can manually move robot
- User can manually activate water spray

5. Relay-Based Water Spraying Logic

A relay module is used to control the water pump system.

When the robot reaches a safe distance from the fire source:

- The relay activates automatically
- Water pump turns ON
- Water is sprayed toward the fire
- Pump turns OFF after extinguishing process

This enables safe electrical control of the water spraying mechanism.

6. ESP32 Integration

ESP32 acts as the main communication and control unit of the robot.

The ESP32:

- Receives commands from laptop
- Controls robot movement
- Controls motors and relay module
- Handles wireless communication
- Executes autonomous commands in real time

The ESP32 integration makes the robot lightweight, wireless, and suitable for real-time robotic applications.

7. Dynamic Navigation and Targeting Logic

The robot continuously tracks the detected fire location using computer vision feedback from the AI model. Based on the fire position, movement commands are dynamically generated and sent to the ESP32 through MQTT communication.

The robot:

- Tracks fire position dynamically
- Adjusts movement in real time

- Moves autonomously toward fire
- Maintains safe stopping distance
- Activates water spray automatically

This creates an intelligent autonomous navigation and targeting system.

Hardware Components

Component	Purpose
Robot Chassis	Base structure of robot
ESP32	Main wireless controller
Phone Camera	Live video feed for computer vision
Relay Module	Controls water pump
DC Motors	Robot movement
Motor Driver L298N	Controls motor direction and speed
Logic Level Shifter	Voltage level conversion
Breadboard	Circuit prototyping
Jumper Wires	Electrical connections
Buck Converter	Voltage regulation
Batteries	Power supply
Water Pump	Fire extinguishing mechanism

Hardware Components and Their Purposes

Software and Tools Used

Software / Tool	Purpose
Python	Simulation and AI logic
Pygame	Visual simulation
VS Code	Development environment
Arduino IDE	Microcontroller programming
MQTT Broker	Communication system
OpenCV	Computer vision
Tinkercad	Circuit simulation

Software Tools and Their Purposes

System Architecture

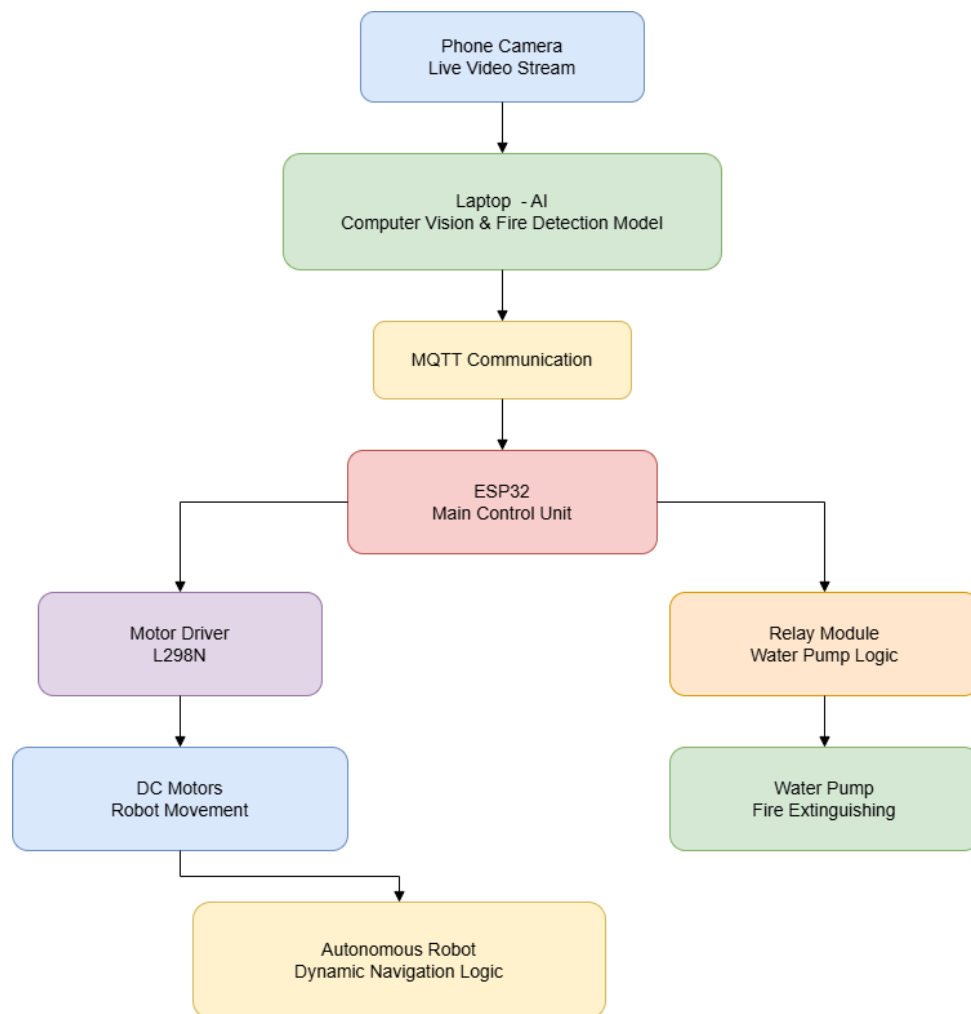
Overall System Flow

Phone Camera → AI Trained Model on Laptop → MQTT Communication → ESP32 → Motor Driver / Relay → Robot Movement & Water Pump

Detailed System Flow

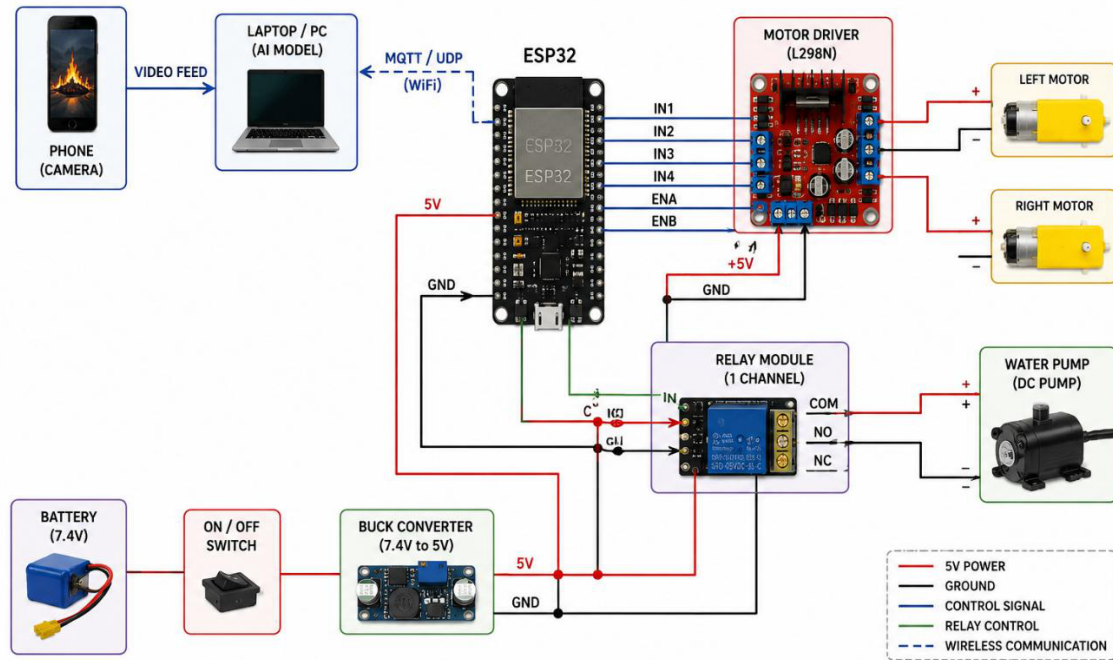
1. The phone camera provides a live video feed of the environment.
2. The laptop receives the video stream and processes it using the trained AI/computer vision model.
3. The AI model detects fire and determines movement decisions dynamically.
4. These movement commands are sent using MQTT communication.
5. ESP32 receives the commands wirelessly.
6. ESP32 controls the motor driver, motors, and relay module.
7. The robot moves toward the detected fire.
8. Once near the fire, ESP32 activates the relay module.
9. The relay turns ON the water pump.
10. Water is sprayed to extinguish the fire.
11. After extinguishing the fire, the robot resumes searching mode.

Block Diagram



System Block Diagram

Circuit Diagram



Circuit Diagram of Captain Obi 2.0

Working / Functionality

The complete operation of Captain Obi 2.0 follows these steps:

1. Robot patrols the environment.
2. Camera continuously scans surroundings.
3. AI model processes live camera feed.
4. Fire is detected using computer vision.
5. MQTT sends movement command to robot.
6. Robot uses dynamic targeting logic to move toward the detected fire.
7. Obstacle avoidance logic activates if needed.
8. Robot reaches safe distance near fire.
9. Water pump activates automatically.
10. Fire gets extinguished.
11. Robot resumes searching mode.
12. If water level becomes low, robot automatically goes to refill station.
13. Robot refills water and resumes operation.

Use-Case Overview

Captain Obi 2.0 is designed to handle different operating situations during fire detection and extinguishing. The robot receives visual input from the phone camera, processes it through the AI model on the laptop, and sends movement commands to the ESP32 through MQTT/UDP communication. Based on the detected condition, the robot decides whether to move, stop, spray water, refill, or continue searching.

Situation / Input	Robot Response
No fire detected	Robot remains in search/patrol mode
Fire detected in camera feed	Laptop AI model sends movement command to ESP32
Fire detected on left/right side	Robot adjusts direction toward fire
Fire detected in front	Robot moves forward toward fire
Robot reaches safe distance	Robot stops and activates water pump
Fire extinguished	Robot stops spraying and returns to search mode
Water level low/empty	Robot moves to refill station
Manual mode selected	User controls robot movement and water spray
Communication command received	ESP32 executes movement or pump action

Use-Case Situations and Robot Responses

Behavior Logic Summary

Captain Obi 2.0 follows a priority-based behavior logic to operate safely and intelligently:

- 1. Computer Vision Detection**

The phone camera captures the environment and sends the video feed to the laptop.

- 2. AI Decision Making**

The trained AI model detects whether fire is present or not.

- 3. Command Communication**

The laptop sends movement or pump commands to ESP32 using MQTT/UDP communication.

- 4. Movement Execution**

ESP32 controls the L298N motor driver to move the robot toward the detected fire.

- 5. Safe Stopping**

The robot stops at a safe distance before activating the water pump.

- 6. Water Spraying**

ESP32 activates the relay module, which turns on the water pump.

- 7. Return to Search Mode**

After extinguishing the fire, the robot resumes searching for another fire source.

Python/Pygame Simulation

A complete software simulation of the project was developed using Python and Pygame.

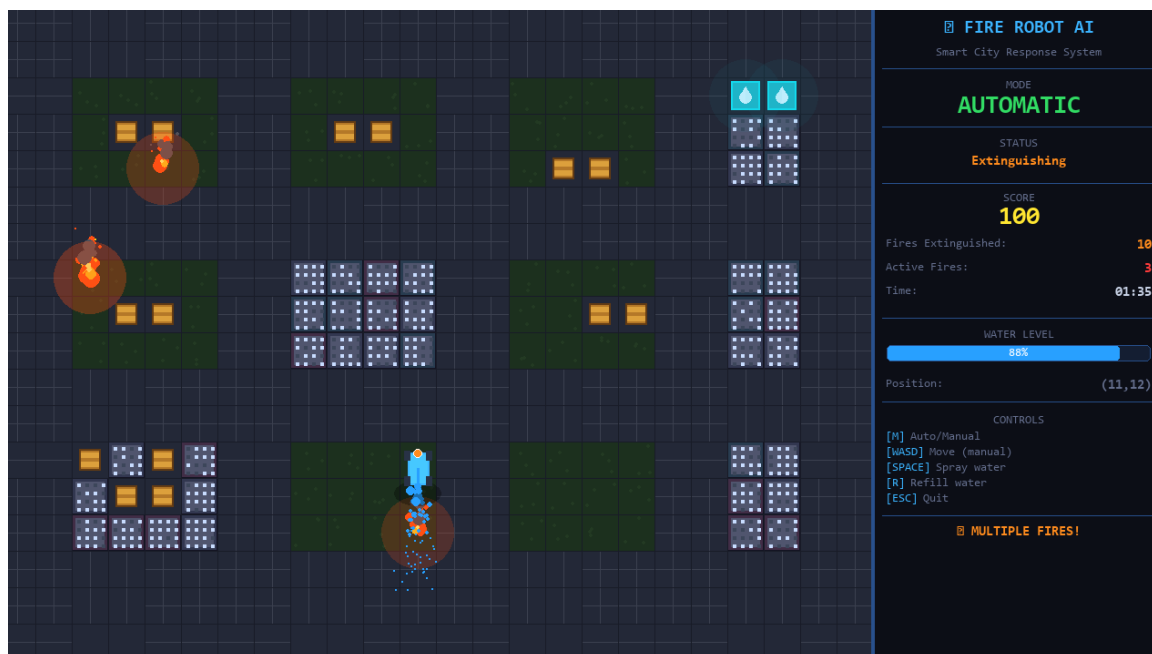
Simulation features include:

- Autonomous mode
- Manual mode
- Dynamic fire spawning
- Water management system
- Refill station
- Dynamic targeting and movement logic
- Obstacle avoidance
- Dashboard and scoring system
- Fire extinguishing animation

The simulation helps test AI logic and robot behavior before physical implementation.

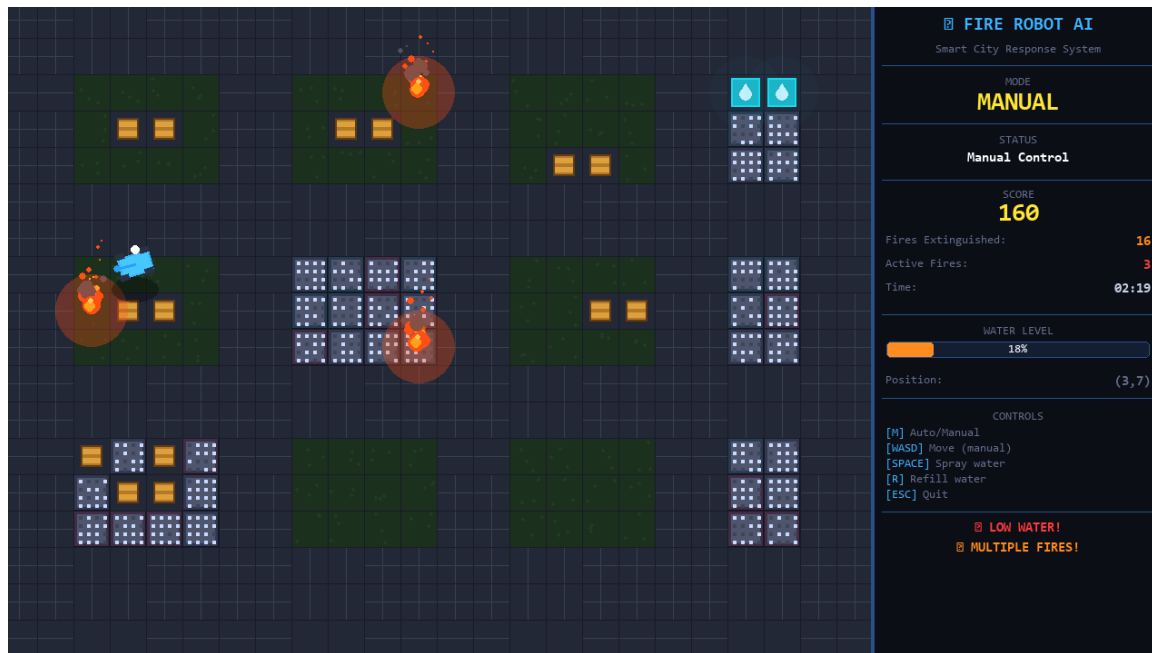
Simulation Screenshots

Automatic Mode



Simulation – Automatic Mode

Manual Mode



Simulation – Manual Mode

Fire Detection



Fire Detection Display

Water Refill Station



Simulation – Water Refill Station

Dashboard and Visualization



Simulation Dashboard and Visualization

Physical Hardware Output

Front View



Robot – Front View



Manual Mode



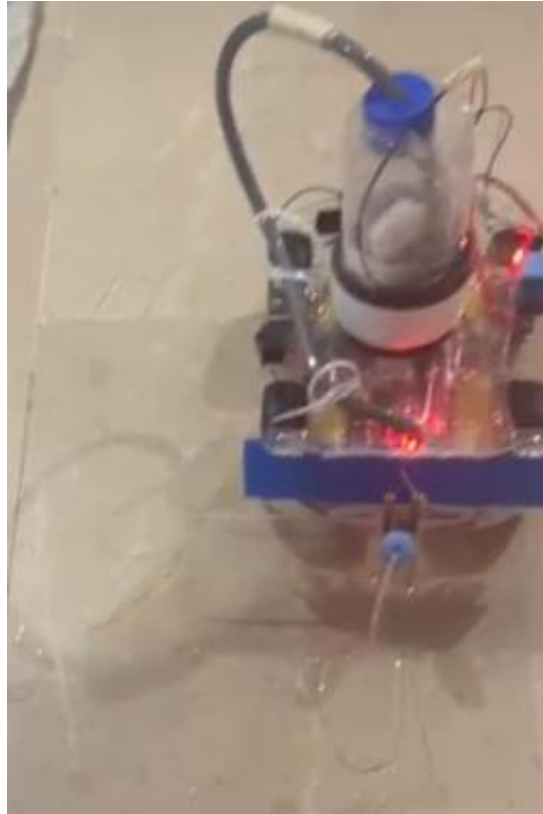
Automatic Mode

Camera Screen View



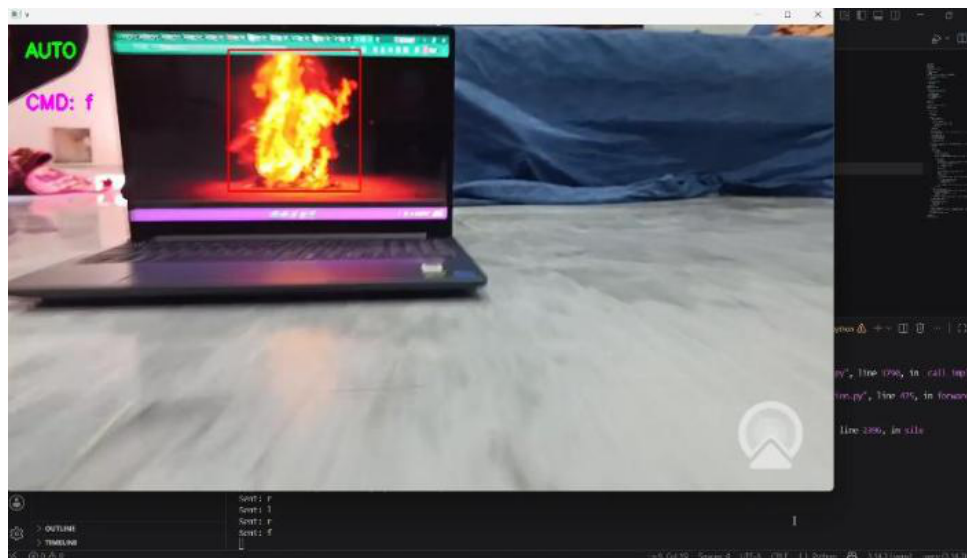
Camera Screen View – Fire Detection in Progress

Water Spraying



Water Spraying Mechanism

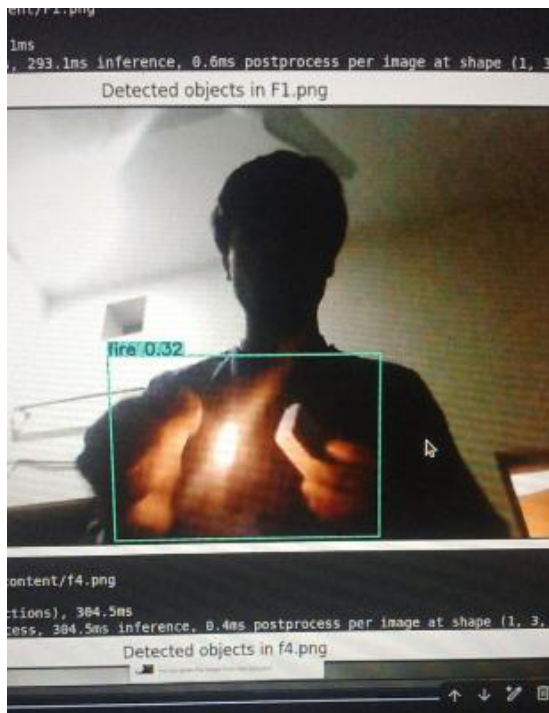
Fire Detection (BTS)



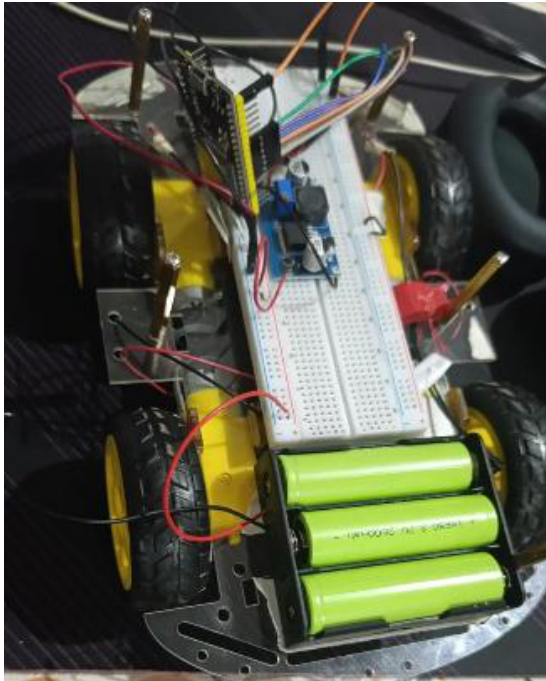
AI Fire Detection on Laptop Screen

Behind The Scenes (BTS)

Team working pictures



Wiring and soldering process

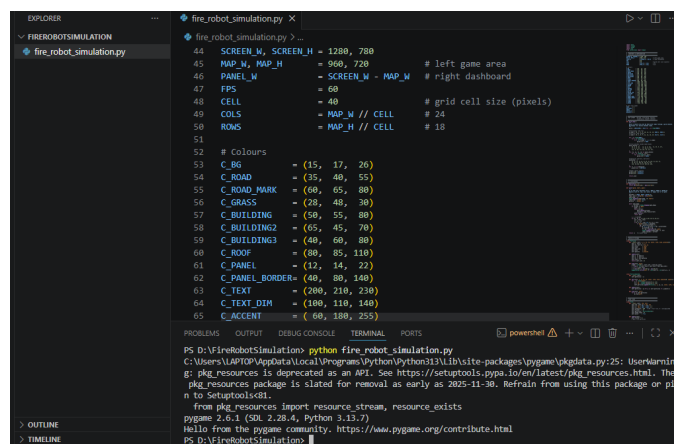
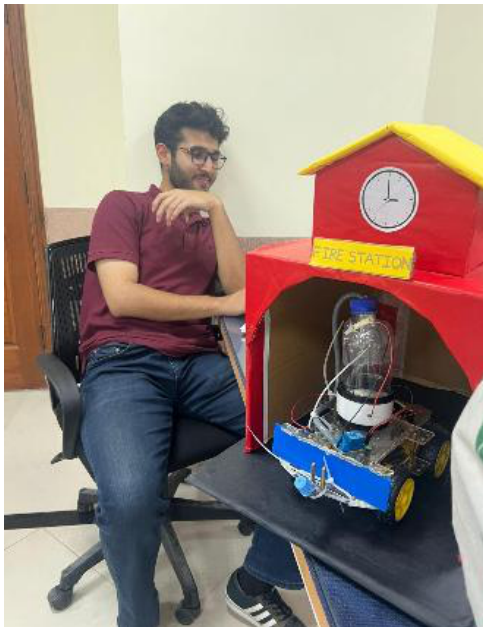


Testing process



System Testing Process

Simulation development screenshots



Simulation Development in VS Code

Work Division Among Team Members

Team Member	Responsibilities
Hafsa	Hardware support, Python/Pygame simulation development, poster design, testing assistance, and project presentation support
Dawar	Hardware integration, project coordination, ESP32 setup, report writing and documentation, testing, and final report compilation
Umar	Hardware wiring, relay and motor driver connections, coding support, debugging, and system testing
Armughan	Hardware support, AI/computer vision support, coding implementation, communication logic, and software integration

Team Member Responsibilities

Conclusion

Captain Obi 2.0 demonstrates how Artificial Intelligence, robotics, computer vision, and autonomous systems can be combined into an intelligent firefighting solution. The project successfully integrates fire detection, autonomous navigation, obstacle avoidance, water spraying, refill management, and simulation into one complete system.

The project highlights the practical application of AI concepts in real-world safety systems and demonstrates the future potential of autonomous robotic emergency response systems.

Complete Code

For codes here is the github repository link. All codes are present here:

<https://github.com/DawarAhmed27/CaptainObi-FireFightingRobot.git>